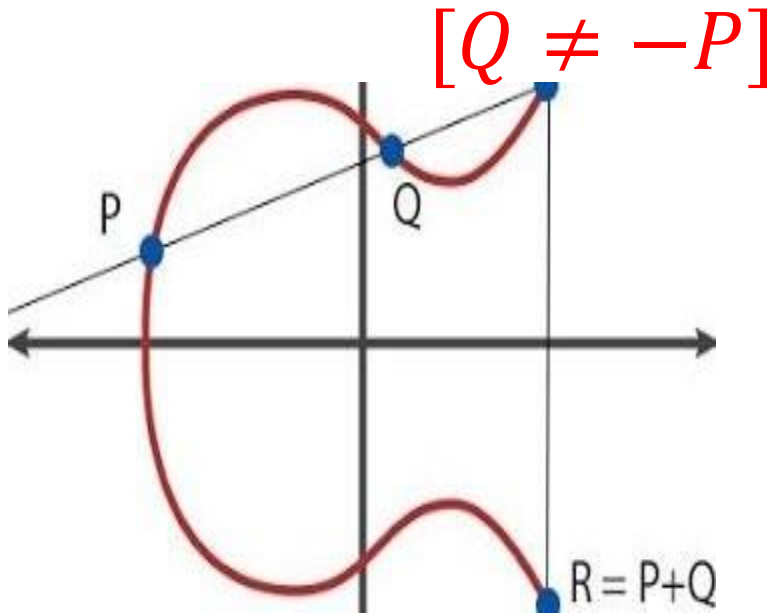


Elliptic Curve Cryptography-2

Module III

Adding two points on the curve

Let the points $P=(x_1,y_1)$ and $Q=(x_2,y_2)$ be in the elliptic group $E_p(a,b)$ and $Q \neq -P$, the rules for addition over the elliptic group $E_p(a,b)$ is,



$$x_3 = \lambda^2 - x_1 - x_2 \pmod{p}$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}$$

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \pmod{p} [P \neq Q]$$

E.g. $E_{11}(1,1)$, $P = (3,8)$, $Q = (6,5)$, $P \neq Q$ and $Q \neq -P$

$$x_3 = 3 \text{ and } y_3 = 3$$

check whether (3,3) is on the given elliptic curve

Adding two points on the curve - example

Consider the field, $F(19)$ and the EC, $y^2 = x^3 + 3x^2 + 1$ over $F(19)$.

List of all the point on the curve.

Check whether $(17,14)$ lies on the above curve

Compute $(8,9) + (12, 13)$